CAPABLE MINDS TUTORIALS (CPM)

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PHY113

PQ

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CPM PHY113 E-QUESTIONS+ Likely E-QUESTIONS

1. Which of the following is not a vector

   A. direction
   B. displacement
   C. momentum
   D. force

2. Golfried libniz's concept of space is a contradiction of which of the following concepts

   A. Immanuel Kant
   B. Albert Einstein
   C. Isaac Newton
   D. Gorge Berkeley

3. A definte amount of physical quantity is called

   A. dimensions
   B. units
   C. S.I
   D. mass

4. Dimension of momentum

   A. $ML^{-2}T^{-2}$
   B. $ML^{-1}T^{-2}$
   C. MLT^{-1}
   D. $ML^{-2}T^{2}$

5. Who discovered space time was curved?

   A. Michael Faraday
   B. Isaac Newton
   C. Albert Einstein
   D. Harman Minskowski

6. Physical space has how many dimensions?

   A. 1
   B. 2
   C. 4
   D. 5

7. Tendency for a body to stay at rest

   A. moment
   B. Inertia
   C. force
   D. momentum

8. When is inertia reference of frame not possible

   A. when the rotation of earth is ignored
   B. when the rotation of the earth is considered
   C. When movement is ignored in the reverse
   D. When water is moving

9. The Dimension nature of an event is

   A. Mass
   B. Time
   C. Length
   D. Interval

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10. Which of the following are derived

i.Area ii. Thrust iii. Pressure

A. none of the above

B. I only

C. ii only

D. i, ii and iii

ANSWERS

1	2	3	4	5
A	C	B	C	C
6	7	8	9	10
C	B	B	B	D

11. According to sir Isaac Newton space is

A. not empirical

B. non-eucliden

C. absolute

D. Flat

12. A comparison of any physical quantity with the standard unit is called

A. Measurement

B. Survey

C. Dimension

D. Measurand

13. A definite amount of a physical quantity is called

A. Measurand

B. Dimension

C. Basic

D. Unit

14. $15x10^3$ J can be written as

A.15KJ

B. 2KJ

C. -1.5KJ

D. 150J

15. What is the unit of Pressure?

A. $ML^{-1}T^{-2}$

B. $ML^{-2}T^{2}$

C. $ML^{2}T^{-2}$

D. $ML^{-2}T^{-1}$

16. What is the unit of Density?

A. $ML^{-2}T^{-2}$

B. $ML^{-3}T^{-2}$

C. ML^{-3}

D. $ML^{-4}T^{-2}$

17. What is newton's first law of motion?

A. A body stays in a state of rest until it is acted upon by an external force

B. for every action there is an opposite reaction.

C. The force of an object is equal to the rate of change of the momentum

D. A body stays in a state of

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rest until it is not acted upon by an external force

18. What is the dimension of energy?

A. $ML^{-1}T^{-2}$

B. $ML^{2}T^{-2}$

C. $ML^{-2}T^{-2}$

D. $ML^{-2}T^{-1}$

19. What is newton's second law of motion?

A. force of an object is equal to the rate of change of the momentum

B. A body stays in a state of rest until it is acted upon by an external force

C. for every action there is an opposite reaction

D. A body stays in a state of rest until it is not acted upon by an external force

20. Einstein's first postulate is?

A. force is equal to the rate of change of momentum

B. The laws of nature are the same in all coordinate systems moving rectilinearly and uniformly relative to each other

C. speed of light is constant

D. speed of light is constant and non-dependent of the speed of the observer

ANSWERS

11	12	13	14	15
C	A	A	C	A
16	17	18	19	20
C	A	B	A	B

21. What is S.I unit?

A. International System of Unit

B. System Unit

C. Systeme Internationale

D. Units of Systems

22. What is the unit of watts?

A. $ML^{1}T^{-2}$

B. $ML^{1}T^{-2}$

C. $ML^{2}T^{-1}$

D. $ML^{2}T^{-2}$

23. What is dimension of Inertia?

A. ML

B. $ML^{2}T^{-2}$

C. M

D. $ML^{2}T^{-3}$

24. A unitless quantity

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A. has non-zero dimension

B. may have non-zero dimensions

C. always have a non-zero dimension

D. never has a non-zero dimension

25. Which of the following quantities has the same dimensions

A. Impulse and Momentum

B. Moment of inertia and torque

C. specific heat and latent heat

D. force and surface tension

26. Checking the correctness of a physical quantity using dimension analysis is based on

A. Types of systems of units

B. method of measurements

C. principle of homogeneity of dimension

D. Equality of frame reference

27. The joule is the unit of which physical quantity

A. force

B. weight

C. power

D. energy

28. Impulse can be defined as?

A. rate of change of momentum

B. change in momentum

C. none

D. product of force and velocity

29. Which of the following is not a vector?

A. Displacement

B. direction

C. momentum

D. force

30. Which of the following cannot be used to measure speed

A. ms^{-1}

B. kms^{-1}

C. mh^{-1}

D. kgs^{-1}

ANSWERS

21	22	23	24	25
C	A	D	D	A
26	27	28	29	30
C	D	A	B	D

31. If a particle moves 2m due east and 2m due west in a straight line. What

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is his displacement?

A. 0

B. 2.59

C. 3

D. 4

32. Which of the following is an example of an object moving in uniform motion?

A. Gas particles

B. A toy car crossing the room at constant speed

C. motor being crashed

D. unbalanced pencil

33. A vector is describe by

A. magnitude and direction only

B. direction only

C. unit, magnitude and direction

D. unit and direction

34. The idea of inertia is postulated by

A. Newton's first law

B. Newton's second law

C. Newton's third law

D. Newton's fourth law

35. Which of the following is not a vector?

A. force

B. direction.

C. momentum

D. displacement

36. Which of the following is not a vector?

A. Displacement

B. acceleration due to gravity.

C. force of gravity

D. work, energy and power

37. Which of the following are scalar quantities?

A. force and gravity

B. work and energy

C. displacement and length

D. frequency and tension

38. Determine the dimension of density,

A. ML^2T^{-1}

B. ML^{-3}

C. ML^2T^{-2}

D. ML^2T^{-3}

39. Find the dimension of power.

A. ML^2T^{-2}

B. ML^{-3}

C. M^3L^2T

D. ML^2T^{-3}

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40. The dimension of momentum is ------,

A. $ML^{-1}T^{-2}$

B. $ML^{2}T^{-2}$

C. MLT^{-1}

D. $ML^{2}T$

ANSWERS

31	32	33	34	35
A	B	A	A	B
36	37	38	39	40
D	B	B	D	C

41. A car of mass 600kg moving with a forward acceleration of 5 m/s^2 is acted upon by a constant resistive force of 1000N. Calculate the force exerted from the engine to maintain this forward acceleration

A. 2KN

B. 4KN

C. 8KN

D. 3KN

42. A car travels 20.0 km due North and then 35.0 km in a direction of 60° West of North. Find the magnitude of the resultant displacement of the car

A. 58.2km

B. 48.2 km

C. 30.0 km

D. 82.5 km

43. A body of mass 2 kg undergoes a constant horizontal acceleration of 5 m/s^2. Calculate the resultant horizontal force acting on the body.

A. 10 N

B. 2.5 N

C. 8N

D. 16N,

44. The major cause of motion is?

A. Gravitational

B. Pull

C. Force

D. Work

45. Which of the following is not a factor that affects motion?

A. Air resistance

B. Viscosity

C. Solid friction

D. Surface tension

46. Which of the following is not an example of circular motion?

A. Car cornering

B. aircraft banking

C. Planetary orbit

D. man walking on a bridge

47. A force acting at 60^0 to the horizontal has an x-component of 45N.What is the value of the force?

A. 110N

B. 90N

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C. 7ON

D. ON

48. A car has an acceleration of 12m/s² if its initial velocity is 10ms-¹ the distance the car covers in 5s is?

A. 15m

B. 25m

C. 53m

D. 20Om

E.

49. A car travels at 20km/h for 1.5h and at 3km/h for 2.0h and at 40kmh⁻¹ for 0.75h. The car average velocity is approximately

A. 19kmh⁻¹

B. 21mh⁻¹

C. 20mh⁻¹

D. 15mh⁻¹

50. A car with its brakes applied has an acceleration of 12m/s² if its initial velocity is 10ms⁻¹ the distance the car covers in 3secs after the acceleration begin is

A. 15km

B. 84m

C. 47m

D. 40m

ANSWERS

41	42	43	44	45
B	B	A	C	D
46	47	48	49	50
D	B	D	D	B

51. A man walks 10km due East and then 20km due South. The displacement of the man from the initial position is

A. 30km South

B. 27.6km 30°N of East

C. 22.36km 63.4° south of East

D. 22.36km 26.57° south of East

52. Calculate the force required to impact an acceleration of 5m/s² to mass of 10kg

A. 50

B. 25,

C. 10

D. 30

53. A gun fires a shell of mass 5kg in horizontal direction. The gun recoils at 0.4m/s and its mass is 3kg. Calculate the velocity of the shell

A. 240

B. 420

C. 42

D. 0.24

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54. Quantities that have both magnitude and direction are called

A. fundamental quantity

B. vector quantity

C. scalar quantity

D. derived quantiy

55. A car weighing 9800N is moving with a speed of 40Km/h. On the application of the brakes, it comes to rest after travelling a distance of 50m. Calculate the average retarding force

A. 1240N

B. 1230N

C. 1209.3N

D. -1240N

56. A 10kg bullet is fired into a 100kg gun with a speed of 1000 m/s. What is the speed of recoil of the gun?

A. 10 m/s

B. 10 m/s

C. 10000 m/s

D. 100 m/s

57. The change in the position of a body is called?

A. Speed

B. Motion

C. Displacement

D. Position

58. The dimension of gravitational constant 'G' is

A. $M^{-1}L^{3}T^{-2}$

B. $ML^{3}T^{-2}$

C. $M^{-1}L^{2}T^{-1}$

D. $ML^{-1}T^{2}$

59. The force that acts on a mass of 1g and gives it an acceleration of $1cms^{-2}$ is defined as

A. Newton

B. Dyne

C. Pound-force

D. Pascal

60. The resultant of two forces X and Y will be greatest when the angle between them is

A. 0°

B. 90°

C. 45°

D. 80°

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ANSWERS

51	52	53	54	55
C	A	D	B	B
56	57	58	59	60
D	C	A	B	A

61. A car accelerates uniformly from rest at 5 m/s². Determine its speed after 10s

A. 0.5m/s

B. 2.0m/s

C. 25.0m/s

D. 50m/s

62. For a projectile the maximum range is obtained a when the angle of projection is

A. 60

B. 30°

C. 45°

D. 50°

63. Which of the following has the same unit as "moment of a force"?

A. Force

B. Power

C. Momentum

D. Work done

64. A stone is projected at an angle of 60° and an initial velocity of 20 m/s. Determine the time of flight.(Take g = 10 m/s²)

A. 34.6s

B. 3.46s

C. 1.75s

D. 17.3s

65. A ball is thrown vertically into the air with an initial velocity 'u'. What is the greatest height reached?

A. u^2/g

B. $3u^2/2g$

C. u^2/g

D. $u^2/2g$

66. Which of the following give the correct dimension for radiant flux?

A. $ML^2T^{-2}K^{-1}$

B. $ML^3T^{-3}K^1$

C. $MLT^{-3}K^{-1}$

D. ML^2T^{-3}

67. The following are instrument for measuring the length of a substance except

A. fluid

B. micrometer screw guage

C. beaker

D. vienier caliper

68. The force which tend to oppose the flow of motion is called

A. center force

B. frictional force

C. opposing force

D. centri-petal force

69. If a force of 40N acting in the direction due east and a force of 3ON, acting in the direction due north.Then the magnitude of the resultant of these two forces would be

A. 40N

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B. 10N

C. 60N

D. 5ON

70. Dimensions of strain are

A. [L]

B. $[M] [L]^{-1} [T]^{-2}$

C. [L]

D. It's a dimensionless quantity

ANSWERS

61	62	63	64	65
D	C	D	B	D
66	67	68	69	70
B	A	B	D	D

71. The following are fundamental quantity EXCEPT,

A. velocity

B. time

C. mass

D. length

72. One of the differences between mass and weight is that__,

A. mass is vector while weight is scalar

B. weight is vector while mass is scalar

C. the unit of mass is Newton while weight is Kilogram

D. both has magnitude and direction

73. Unit that cannot be derived from any other unit are called

A. rare

B. units

C. basics unit

D. fundamental unit

74. The change in the position of a body is called

A. Speed

B. Velocity

C. Motion

D. Acceleration

75. The property of a body to remain at rest or to remain in motion with constant velocity is called

A. inertia

B. momentum

C. force

D. force stop

76. A body of mass 5kg move with a velocity of 5m/s for 5s. What force has the body applied

A. 5N

B. B.10N

C. 15N

D. 20N

77. Quantities that have both

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magnitude and direction are called

A. fundamental quantity

B. vector quantity

C. scalar quantity

D. basic quantity

78. The branch of mechanics that deals with particles at rest or in equilibrium is called

A. Kinematics

B. Dynamics

C. Statics

D. Gravitation

79. What is the primary concept introduced by Einstein's theory of General Relativity?

A. Quantum Mechanics

B. Dark Matter

C. Space-Time Curvature

D. Atomic Structure

80. According to Special Relativity, how does time behave for an object moving at a high speed?

A. Time speeds up

B. Time slows down

C. Time remains constant

D. Time becomes unpredictable

ANSWERS

71	72	73	74	75
A	B	D	A	A
76	77	78	79	80
A	B	C	C	B

81. What is the dimension of volume?

A. L^2

B. L^{-3}

C. L^3

D. $L^2 + L^1$

82. What does the theory of General Relativity explain about gravity?**

A. It's a magnetic force

B. It's a force between masses

C. It's the curvature of space-time caused by mass

D. It's a repulsive force

83. In a space-time diagram, what do straight lines represent?**

A. Paths of light

B. Paths of objects at rest

C. Paths of objects in uniform motion

D. Paths of objects under gravity

84. What is the name for a region in

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space-time where gravity is so strong that nothing, not even light, can escape its pull?**

A. Solar System

B. Nebula

C. Black Hole

D. White Dwarf

85. How does Special Relativity explain the constancy of the speed of light?

A. The speed of light changes based on the observer's motion

B. The speed of light depends on the observer's location

C. The speed of light is determined by the medium it travels through

D. The speed of light is the same for all observers, regardless of their motion

ANSWERS

81	82	83	84	85
C	C	C	C	B

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